Multi-Agent Physics-Informed Neural Network with MARL

Architecture Overview (1,802,037 parameters)

Phase 1a: **Decoupled** data — Phases 1b–3: **Coupled** data (10% calibration)

MRST
Flow Simulator
$\phi,\ k,\ z,\ t$
19 feat | $n = 625\,\mathrm{k}$

PHREEQC
Geochemistry
$T,\ \mathrm{pH},\ I,\ E_h,\ \eta_\mu,\ \eta_m$
8 feat | $n = 1\,\mathrm{k}$ (LHS)

Brooks-Corey
Hysteresis
$f_d,\ \lambda_{\mathrm{BC}},\ h_{\mathrm{hist}}$
3 feat | $n = 20\,\mathrm{k}$

PINN-Hydro (H)
4 ResBlocks + Attention
$\mathbb{R}^{19} \to \mathbb{R}^3$
907,656 params

$m_{H\to G}$

Geochem (G)
3-layer MLP
$\mathbb{R}^{8} \to \mathbb{R}^5$
35,333 params

$m_{G\to Y}$

Hysteresis (Y)
3-layer MLP
$\mathbb{R}^{3} \to \mathbb{R}^2$
34,306 params

Per-agent reward: $R_i = R_i^{\mathrm{phys}} + \gamma\, R^{\mathrm{collab}} \qquad (\gamma = 0.3)$

H→G→Y→H

$m_{Y\to H}$

Gated Communication

$$m_{i\to j} = \sigma(W_g[h_j\,;\,W_p h_i]) \odot W_p h_i, \quad d_m = 32$$

Hidden-State Summary Projections

$$\tilde{h}_i = W_i^{\mathrm{proj}} h_i, \qquad \tilde{h} = [\tilde{h}_H;\ \tilde{h}_G;\ \tilde{h}_Y] \in \mathbb{R}^{96}$$

$256 \to 32,\ 128 \to 32,\ 128 \to 32$ (concat.)

Attention-Based Consensus Layer

$$\hat{y} = \mathrm{Dec}\left(f + \textstyle\sum_i \alpha_i V_i\right)$$

$$\alpha_i = \mathrm{softmax}\left(\frac{QK_i^\top}{\sqrt{d}} + \lambda\, c_i\right), \quad c_i = 1/\sigma_i^2$$

Input: 14-dim (agent $\oplus$ flow) + 96-dim $\tilde{h}$ (additive, zero-init) | 632,198 params

Physics Constraint Layer

$$S_w + S_g \le 1 \qquad P > 0 \qquad \nabla\cdot\mathbf{v} \approx 0$$

Predictions with Uncertainty Quantification

$$\hat{P} \pm \sigma_P \qquad \hat{S}_w \pm \sigma_{S_w} \qquad \hat{S}_g \pm \sigma_{S_g}$$

$$\sigma^2_{\mathrm{total}} = \sigma^2_{\mathrm{epistemic}} + \sigma^2_{\mathrm{aleatoric}}$$

Hydro | Geochem | Hysteresis | Comm. | Consensus | Output